Problem 1: Mall Customer Segmentation

Dataset: Mall Customer Segmentation

Questions:

- Why do you need to scale the features before applying K-Means?
- Use the Elbow method (K=1 to 10). What is the optimal number of clusters?
- Train K-Means with K=5. How many customers are in each cluster?
- What are the cluster center coordinates for K=5?
- Which cluster would you target for luxury products and why?

Problem 2: Iris Dataset Clustering

Dataset: Iris Dataset

Questions:

- Scale the 4 features. Why is scaling necessary?
- Use the Elbow method and Silhouette Score to find optimal K. What is the best K?
- Train K-Means with K=3. How many samples in each cluster?
- Compare K-Means labels with actual species. Which cluster matches Setosa?

Problem 3: Credit Card Customer Segmentation

Dataset: Credit Card Dataset

Questions:

- Which columns have missing values? How will you handle them?
- Use the Elbow method and Silhouette Score. What is the optimal K?
- Train K-Means with K=4. What are the cluster sizes?
- What are the average BALANCE, PURCHASES, CREDIT_LIMIT, and PAYMENTS for each cluster?
- Which cluster is most profitable for the bank? Which is risky?

Problem 4: Anomaly Detection with K-Means

Dataset: Credit Card Fraud Detection

Questions:

- What is the fraud percentage in this dataset?
- Train K-Means with K=10. For each point, calculate distance to nearest cluster center.
- Set a threshold at the 95th percentile of distances. How many frauds are detected?
- What is the precision and recall for fraud detection using this method?
- What are the limitations of using K-Means for anomaly detection?